

Alex Rivera

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Chennai, India

EDUCATION

Riverside Institute of Technology, Chennai

Bachelor of Technology, Computer Science & Systems Engineering

Sep 2023 – May 2027

EXPERIENCE

Helix Dynamics

Jun 2025 – Jul 2025

Project Intern

Chennai, India

- Reduced message delivery latency to <100ms by optimizing WebSocket payload serialization and implementing per-room connection pooling, while maintaining AES-256-GCM + RSA-OAEP E2E encryption with zero server-side key exposure
- Containerized Node.js WebSocket server and Express API with Docker Compose; wrote 15+ integration tests covering connection drops, key exchange failures, and message replay attacks

Skilstack Academy

Feb 2026 – Apr 2026

Python Intern

Remote

- Engineered PyKV, an in-memory LRU cache engine with O(1) GET/SET via OrderedDict + asyncio.Lock; implemented AOF append-only persistence with atomic log compaction, reducing disk usage by 60% over 30-day operation
- Built automated benchmarking pipeline with 20-way concurrent httpx load, characterizing true server throughput at 708 ops/s (1.41ms/op) vs. in-process dict baseline; identified serialization and connection pooling as primary latency drivers

PROJECTS

CodeSensei | TypeScript, AWS, Bedrock

- Architected 4-layer Socratic reasoning pipeline (Beginner → Design → Pitfalls → Socratic) via structured JSON prompts to Amazon Bedrock Nova Lite; multi-turn conversation history with role alternation prevents context loss across follow-ups
- Optimized cross-region architecture (ap-south-2 Lambda → us-east-1 Bedrock) reducing model costs by 60% while maintaining sub-2s end-to-end latency; DynamoDB TTL enables 30-day per-user history at INR 0 idle cost
- Built auto-explain provider with 1.5s diagnostic debounce and CodeLens coverage for 10+ languages; serverless stack (SAM, Lambda, API Gateway) scales to zero with INR 0/month when inactive

WatchDog | Python, SQLite

- Built user-space background daemon implementing async producer-consumer pipeline with queue.Queue and SQLite executemany batch writes, eliminating synchronous I/O blocking on telemetry ingestion
- Replaced static threat thresholds with Exponential Moving Average (EMA) algorithm learning per-process resource baselines; flags anomalies at 3σ deviation with <5% false positive rate during 72-hour testing
- Enforced tamper-evident log integrity via canonical JSON serialization + SHA-256 hashing; implemented session-authenticated glassmorphic SOC dashboard with real-time threat feed and forensic audit trail

ACHIEVEMENTS & AWARDS

OpenForge '26 Finalist | OpenForge Foundation

May 2026

- Selected for the final round after the proposal stage.

HackFusion '25 Winner | HackFusion

Nov 2025

- 1st place among 100+ teams.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, JavaScript, C++, Java, SQL

Frameworks: Tauri, React, FastAPI, Node.js, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, SQLite, DynamoDB

Cloud: AWS, Lambda, Bedrock

DevOps: Docker, Git, GitHub Actions, Linux

Tools: NumPy, Pandas